Homework 2

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Exercise 1

The action profile $(e, \ldots, e)$ is a Nash equilibrium for all values of $e \in \{1, \ldots, K\}$. In this profile, a player's payoff is $2e - e = e$. For any individual player, decreasing their effort to e_i would result in a payoff of $2e_i - e_i = e_i < e$, and increasing their effort would result in a payoff of $2e - e_i = e + (e - e_i) < e$.

Let $(e_1, \ldots, e_n)$ be an action profile where not all of the effort levels are the same. Let e^* be the minimum, and let some player k have an effort level exceeding e^*. Their payoff is $2e^* - e_k < e^*$, while if they deviate to e^*, their payoff if $2e^* - e^* = e^*$, so the profile is not a Nash equilibrium.

Exercise 2

The action profile (k, k, k) is not a Nash equilibrium for $k \geq 2$, and is for $k = 1$. Their payoff is $1/3$, and if they deviate to $k-1$, $k-1$ will be closer to $\frac{2}{3} \cdot \frac{(k+k+(k-1))}{3} = \frac{2}{3}k - \frac{2}{9}$. The only value for which the profile is a Nash equilibrium, $k = 1$, is a Nah equilibrium because deviating lower is no longer possible.

Let (k_1, k_2, k_3) be an action profile where not all three integers are the same. Let $k^* = \max(k_1, k_2, k_3)$. Suppose only one person has k^* as heir integer, and without loss of generality let it be $k^* = k_3$. Then $\frac{2}{3} \cdot \frac{k_1+k_2+k^*}{3} = \frac{2}{9}(k_1 + k_2 + k^*)$. If $k_1 \geq 29(k_1 + k_2 + k^*)$, k^* does not win since $k^* > k_1$. Otherwise, $k^* - 29(k_1 + k_2 + k^*) = \frac{7}{9}k^* - \frac{2}{9}k_1 - \frac{2}{9}k_2$, while $29(k_1 + k_2 + k^*) - k_1 = \frac{2}{9}k^* - \frac{7}{9}k_1 + \frac{2}{9}k_2$, which is smaller, and so k^* still does not win. In either case, they would be better off decreasing their integer, so the profile is not a Nash equilibrium. Now instead suppose two people have k^* as their integer, and without loss of generality let it be k_2 and k_3. Then $\frac{2}{3} \cdot \frac{k_1+k^*+k^*}{3} = \frac{2}{9}k_1 + \frac{4}{9}k^*$. Then, since $\frac{2}{9}k_1 + \frac{4}{9}k^* < \frac{1}{2}(k^* + k)$, k^* dos not win. They would be better off decreasing their integer, so the profile is not a Nash equilibrium.

Exercise 3

Part (a)

We can model selfish behavior by the following strategic game:

	Sit	$Stand$
Sit	$1,1$	$2,0$
$Stand$	$0,2$	$0,0$

In this game, player 1 would rank the outcomes as following:

$$(Sit, Stand) > (Sit, Sit) > (Stand, Sit) = (Stand, Stand)$$

This is not equivalent to the prisoner's dilemma, regardless of how the actions are named. The Nash Equilibrium is (Sit, Sit).

Part (b)

We can model altruistic behavior by the following strategic game:

	Sit	$Stand$
Sit	$2,2$	$0,3$
$Stand$	$3,0$	$1,1$

In this game, player 1 would rank the outcomes as following:

$$(Stand, Sit) > (Sit, Sit) > (Stand, Stand) > (Sit, Stand)$$

This is equivalent to the prisoner's dilemma, with $Stand = C$ and $Sit = N$. The Nash Equilibrium is $(Stand, Stand)$.

Part (c)

In both equilibria, both people are acting accoridn got their selfish preferences.

Exercise 4

Part (a)

Suppose h hunters pursue the stag. If $h \geq m$, the hunters who did not pursue the stag are better off pursuing the stag, since they prefer $1/n$ of the stag to the hare. If $h < m$, they are better off not pursuing the stag, since they prefer a hare to nothing The equilibria are then $(Stag, \ldots, Stag)$ and $(Hare, \ldots, Hare)$.

Part (b)

Suppose h hunters pursue the stag. If $h > k$, the hunters who did pursue the stag are better off not pursuing the stag, since they prefer the hare to $1/k$ of the stag. If $k > h \geq m$, the hunters who did not pursue the stag are better off pursuing the stag, since they prefer $1/k$ of the stag to the hare. If $h = k$, the profile is a Nash equilibrium. If $h < m$, they are better off not pursuing the stag, since they prefer a hare to nothing The equilibria are then the profile where exactly k hunters pursue the stag and $(Hare, \ldots, Hare)$.

Part (c)

The presence of a church means that hunters now prefer $\frac{0.9}{m}$ of a stag to a hare, but prefers a hare to $\frac{0.9}{m+1}$ of a stag. This is the same as the origina stag hunt, but with $m' = \frac{m}{0.9} = \frac{10}{9}m$. Effectively, it just raises the threshold number of hunters that makes stag hunting preferable to hare hunting.